

# Airbag-Based Protective System for UAV Medical Payloads

Design and Simulation in LS-DYNA

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## Design and Simulation in LS-DYNA

by

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# Preface

*A preface...*

*Annamaria Iannuzzo  
Delft, October 2026*

# Summary

*A summary...*

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# Nomenclature

*If a nomenclature is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.*

## Abbreviations

| Abbreviation | Definition                        |
|--------------|-----------------------------------|
| ISA          | International Standard Atmosphere |
| ...          |                                   |

## Symbols

| Symbol | Definition | Unit                 |
|--------|------------|----------------------|
| $V$    | Velocity   | [m/s]                |
| ...    |            |                      |
| $\rho$ | Density    | [kg/m <sup>3</sup> ] |
| ...    |            |                      |

# Introduction

## 1.1. Motivation

The present MSc Thesis in Aerospace Structures and Materials falls under the category of *Engineering Design*, in alternative to the *Scientific Research*-type projects. In *design* projects, it is required to attempt the process of innovation not through the discovery and exploration of a research niche, but rather by applying pre-existing scientific knowledge to enable the concrete realisation of a technical design. In the case of the present thesis, the overall objective consists in proposing a *design workflow* and validating it through testing and simulation. If adopted in the preliminary design phase of a drone-based delivery mission, this workflow yields an optimised airbag-type device tailored to a specific failure-and-drop application, in line with the requirements set by the project commissioners. Therefore, the successful completion of this thesis is achieved by:

1. Proposing an application-specific design workflow that fits within the broader context of the ongoing research effort;
2. Testing the workflow, both experimentally and through simulation, to realistically inform the device's design.

The design workflow (informally called *inner (design) loop* throughout this thesis) is intended as a first attempt to bridge the gap between the long-term, broadly-defined objective of constructing a large-scale drone network (also called *outer (design) loop*) over the region of Piedmont, Italy, and the previously completed research work, consisting in a series of logistic, legislative, and generally high-level evaluations of possible **drone-based missions** with the objective to **deliver medical supplies in emergency scenarios**. Such work was conducted *Politecnico di Torino*, in collaboration with the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex and automotive consulting company *FEV Italia S.r.l.*, three of the main industrial and research stakeholders working towards helping hospitals in the transition from road to air transport to enhance the logistics of the *Sistema Sanitario Nazionale (SSN)*, the Italian National Healthcare System) both at an intra- and inter-hospital level.

This MSc Thesis Project is, therefore, a multidisciplinary project where the knowledge and techniques of Structural Engineering are put to the service of the medical field. The airbag-like device resulting from the design process should be able to protect the free-falling medical payload in the event of a drone failure and its subsequent detachment from the latter. The device aims to bring the impact accelerations within the allowable limits of the payload container and of the medical payload being transported, in compliance with the safety regulations regarding dispersion of chemical substances into the environment. However, contrary to most aeronautical projects which do not distinguish between the integrity of the aerial vehicle and the safety of its passengers, this project adopts a "payload-first" approach where the protection of the highly valuable medical cargo is prioritised over the integrity of the drone carrying it. Distinguishing between the safety aspects of the unmanned aerial vehicle (UAV) and its payload implies a certain facilitation in the certification process of the UAV, as well as for the modular implementation of the safety device on a variety of UAVs serving in different, specific delivery

missions.

## 1.2. Project framing

### 1.2.1. Care centralisation and territorial inequity in the Italian healthcare system

The idea of transforming healthcare delivery networks from road- to air-transport-based systems stems from the consensus within the Italian medical field, consolidated during the public health emergency of the COVID-19 pandemic (2020–2022), that the current ambulance-based architecture of the SSN's logistic infrastructure (described in detail<sup>1</sup> in [1]) cannot accommodate the healthcare system's needs and transitions. The two main - and competing - factors are:

1. **Care centralisation:** [2] describes care centralisation as the ongoing process of concentrating medical expertise and services in key geographical areas, e.g. provincial and/or regional capitals, to meet the demands of high-volume care. The downside of care centralisation is that at a regional level, it induces "distance-related inequity, particularly for rural and peripheral populations", especially heightened in emergency scenarios.
2. **Unhomogeneous distribution of emergency services :** [3] affirms that, in contrast with the centralisation manouver described above, emergency resources are still unhomogeneously distributed over the regional territories, with services overloaded in densely populated areas and underused in low-density ones, resulting in inefficient overall utilisation.

In conclusion, [4] builds on these two points by affirming the importance of establishing a generally more efficient strategy for healthcare management at all levels, and incentivises the development of technologies suitable for upgrading, among the several aspects of the SSN, the logistics of the Italian healthcare supply chain.

### 1.2.2. Prior research and positioning of the present thesis

In response to the issues highlighted in Section 1.2.1, recent years have seen an increasing number of technology development initiatives where industry players in the transport sector and research institutions work together to push for the modernisation of transportation means within the Italian SSN. Several of these industry-research partnerships have ventured into the globally booming market of unmanned aerial vehicles, where remotely-piloted or fully autonomous drones are being used ever more often to carry medical payloads in a plethora of scenarios: intra- vs. inter-hospital delivery [5], ordinary logistics vs. emergency missions and humanitarian applications[6], experimental/direct flight between strategic poles [7] vs. a fully-operational, inter-hospital, hub-based UAV network covering national territory (the *Zipline* case study[8]).

Following the global trend to answer a local need, in the specific case of the Piedmont region in north-west Italy, where at the time of writing there exists no city- or region-wide UAV network deployed for the delivery of medical goods, many pilot programmes (further analysed in Section 2.1) are emerging, among which a partnership between the Pharmacy department of the *A.O.U. Città della Salute e della Scienza di Torino* hospital complex (head-quartered at Molinette Hospital in Torino), a research group within the Department of Mechanical and Aerospace Engineering at *Politecnico di Torino*, and the Numerical Anaysis Team of *FEV Italia S.r.l.*, the Italian branch of the German-founded multinational company *FEV*, a global leader in research and development for the automotive industry.

The seven-year-long partnership between these three institutions is the backbone of the present MSc thesis project, the most recent of a chain of research projects aiming to pursue the long-term vision of flanking, if not completely replacing, the current road-based transportation means with aerial alternatives within hospitals in Piedmont and, specifically, in the city of Torino (the region's capital), in favour of more rapid, more efficient, and safer transportation of medical supplies. Specifically, this research effort has broadly tackled the key aspects required to establish a complete and regulation-compliant UAV medical delivery ecosystem in the Piedmontese territory, resulting in five MSc theses and one conference paper presenting the following results:

<sup>1</sup>"The type of vehicles used varies from region to region. In Piedmont it mainly consists of ambulances used only for transport plus more advanced vehicles equipped for resuscitation, replaced in some sites by mSa on a car with a doctor, nurse and rescuer driver, organized on the basis of specific needs of the territory, different in the four macro-areas of the emergency network (rEmE) in which the region was divided." [1]

- **Cavallo (2020)**[9]: Preliminary assessment of UAV platforms suitable under Italian and European regulations. Development of risk maps and definition of optimal trajectories between selected hospitals.
- **Bertelli (2021)**[10]: Detailed selection of UAV based on mission specifications derived from the medical goods to be transported and ENAC regulatory constraints. Application of the SORA (Specific Operations Risk Assessment) methodology to three feasible mission routes, yielding the safety requirements that operators must satisfy.
- **Sivieri (2021)**[11]: Identification and characterisation of thermal and vibrational loads encountered during typical UAV medical missions. Experimental testing on milk-derived sera samples to determine their resistance to UAV-induced vibrational stresses. First hypothesis of medical payload container geometry and materials, implemented in the experimental campaign.
- **Fiorentino (2021)**[12]: Development of a path-planning strategy for life-saving drug distribution in populated environments. Generation of risk maps and implementation of an RRT-based algorithm to minimise global risk and flight time.
- **Angelini (2023)**[13] and **Angelini (2024)**[14]: Comparative analysis of the safety and economic efficiency of emergency medical deliveries performed by car, drone, and VTOL aircraft. Identification of optimal pharmaceutical delivery hub locations for each transport mode to maximise logistical performance.

The present thesis was carried out in collaboration with the above-mentioned partners to advance the research effort, bridging the gap between the long-term objective and the results achieved so far. This required preliminary work aimed at formalising the project requirements at both a broad, high level and a specific, first-step level. Having taken a legislative, logistic, market and mission-evaluation approach until now, it was possible to delineate a scenario in which the approval of *ENAC* (the Italian Civil Aviation Authority - *Ente Nazionale per l'Aviazione Civile*) could potentially be obtained to enable this type of operation. However, in a scenario where certifications are specific to mission type, trajectory and vehicle, it appeared necessary to implement a more concrete approach rooted in *design*, whilst adopting a "payload-first" approach that diverted the project towards a device for the *safety of the medical payload* rather than of the aerial vehicle. The deliberate prioritisation of this aspect is justified in Sections 2.1 and 2.2, where it is illustrated on the basis of a regional market analysis and exploring a paradigm shift in the design process of biomedical devices.

### 1.2.3. Stakeholders

Overall, the stakeholders participating in the present project had the following roles:

- **TU Delft** (Prof. Bianca Giovanardi): Primary academic supervisor overseeing the scientific quality and methodological development of the research.
- **Politecnico di Torino** (Prof. Enrico Cestino, Davide Angelini): Institutional leads and co-supervisors for the broader UAV medical-delivery research project, providing access to the experimental platforms used.
- **A.O.U. Città della Salute e della Scienza di Torino / Molinette Hospital**: Provider of medical and operational data related to pharmaceutical transport requirements and mission scenarios.
- **FEV Italia S.r.l.**: Industrial partner offering technical support for advanced modelling and simulation activities, including SolidWorks-based design and LS-DYNA impact analyses.

The interaction between stakeholders, as well as the management and communication aspect of the project, is analysed in Chapter 6 - "Project Management".

## 1.3. Document Outline

DISCORSIVO. Esempio:

The thesis is structured as follows, first, chapter 2 presents a literature review to outline the several possibilities to predict the buckling load of cylindrical shells and which techniques are available to reduce the sensitivity to imperfections. The chapter concludes by identifying the research gaps related to the buckling of cylindrical shells. From this conclusion, the research objective and scope are defined

in chapter 3. The goal is to include imperfections within the displacement-based Koiter methodology and the necessary steps to achieve this are explained in chapter 4 to 7, as well as how effective this novel modelling technique is. Finally, the concluding remarks can be found in chapter 8 and future research is presented in chapter 9.

# 2

## State of the Art and Research and Development Gap

(BACKGROUND KNOWLEDGE) - put a little introduction paragraph here, saying that this chapter gathers important information found in literature and functions as a literature research, tailored to answering the problem, exploring the market, understanding airbag design criteria also through the historical perspective, and then producing the main research questions.

### **2.1. first reason: mini-market analysis - the technology in piedmont**

\* ab-zero \* SkyEU \* us

\* mention CROCE ROSSA ITALIANA (?necessary here? maybe earlier)- launched initiative delivery (ref. 6, tesi sivieri)

\* It is evident from research that every solution is possible, the challenge is to have normative approval rather than creativity.

### **2.2. second reason: The payload-first approach**

Chhetri - an updated REVIEW of medical drones in 2026 \* overcome geographical barriers and provide critical care in remote or disaster-stricken areas. \* critical role of payload container design, including fixed, cable-suspended, and temperature controlled systems, and analyze its effect on both the medical supplies and the overall flight dynamics of the aircraft. \* These supplies may include vaccines, medications, blood units, diagnostic samples, and organs for transplantation.

regarding the payload container: "It must also meet safety standards, ensuring dust tight sealing and protection against strong water jets, thus shielding the container and its contents from mechanical impacts and external forces. Aerodynamic efficiency is another vital consideration as it lessens the power necessary for flight. Additionally, the choice of materials is crucial to the container's performance, with lightweight yet sturdy materials selected to preserve structural integrity while preventing unnecessary energy use. Finally, the transport container must be operationally compatible, providing flexibility in deployment across various aerial transport platforms." does not mention protection of what is inside.  
— CHECK REFERENCE 25

\* drone is designed first, then "put a parachute on it" (the safety requirement by Alulema, which Mass feedback loop: a heavier aircraft needs a bigger parachute, which makes it heavier. This is why the requirement costs 36 percentage points of feasibility (Table 1: 84\* focus on vibration isolation and temperature isolation, keeping the cold chain \* general impression: that the impact on aerodynamics and flight mechanics is included before even thinking about the integrity of the payload in case of impact.

WHAT IF WE THOUGHT ABOUT REVERSING THE DESIGN PROCESS? Design the drone AFTER

having hypothesised the safety device that is included in the payload container in the first place? QUOTES: "Success is measured at the patient, not at the device." Success is measured at the vial, not at the airframe. A drone that survives while the phial arrives cracked has failed the mission. The same goes for a device that works but doesn't serve the patient.

Rodriguez Drawing a parallel to the medical tech design approach, which is often something that medical tech designers need to be reminded of when designing the coolest tech for patients. "A shift in the traditional technocentric view of medical device design to a human-centered one is needed to bridge existing translational gaps and improve health equity. called "translation gap" or "implementation gap"

in this case, the gap is between the technological potential (the drones) and 2 ends, i.e. the normative safety constraints where a parachute is not enough, and, most importantly, protecting the payload which is arguably more valuable than the drone itself.

## 2.3. What I should know for airbag design

### 2.3.1. Historical perspective

\* Automotive Engineering

\* Space Engineering - implemented in impact applications, similar to the one requested for this case, since the 1960's for xxx mission. - analytical, hard-codable solutions are 1Dof and rare, FEM is immediately chosen to include 2- and 3d effects of impact.

however, both fields have the guidelines in common, which are explained in the following paragraph

### 2.3.2. Design guidelines

\* Avendo un panorama più chiaro su quella che è l'obiettivo a Torino, si è scelto di concentrarsi sulla mitigazione del ground risk whatever. La scelta ricade su una tipologia di dispositivo di sicurezza atto a mitigare il rischio di impatto, danneggiamento delle fiale, dispersione nell'ambiente, etc. un Airbag

Perché è stato scelto l'airbag? paracadute: \* pesanti (spessi) \* non reggono impatto \* paracadute è soltanto sul drone - medicinali ad ogni modo persi \* sgancio + paracadute meglio di sgancio + airbag — obiettivo, protezione medicinali, anche se il drone viene protetto tanto meglio \* vantaggi: può essere usato in ospedale (basse quote) \* in acqua ha più senso impermeabilizzare l'airbag e non il contenitore, con il vantaggio che può stare effettivamente a galla

Stabilito che l'airbag è la tecnologia su cui ci vogliamo concentrare, allora le cose da prendere in considerazione sono: lista delle cose del documento in cui ho preso appunti.

Variabili da prendere in considerazione quando c'è da progettare un airbag

Phase 1 variables for peak acceleration 1. Shape 2. Size 3. Fabric 4. Vent size 5. Vent burst pressure 6. Inflation pressure

Phase 2 variables for deployment time 1. Housing/container size and geometry 2. Air injection inlet pressure 3. Air injection mass flow rate 4. Seams locations 5. Folding pattern 6. Number of folds

## 2.4. Research Objective

Building on the findings of previous theses and scientific works (especially those related to mission specifications, regulatory constraints, and environmental loading), the present research aims to:

Design, model, and simulate an airbag-based protective system for UAV medical payloads using LS-DYNA, with the objective of ensuring the structural integrity of pharmaceutical materials and their containment in the event of drone failure (including parachute) and emergency payload release, therefore preventing environmental dispersion and guaranteeing civil safety.

Specifically, the project will:

- Identify and characterise the standard payload container's mechanical allowables based on general packaging requirements of the medical goods intended for UAV transport.
- Develop a conceptual and numerical design of an airbag system capable of deploying during emergency release conditions to absorb impact energy and maintain payload integrity.
- Simulate the dynamic behaviour of the airbag and payload in LS-DYNA under realistic worst-case impact scenarios.
- Assess performance metrics, such as deceleration limits, airbag deployment time, and pharmaceutical survivability.

How can a systematic multi-phase simulation workflow be developed to design an airbag-based protective system for medical payloads during UAV emergency-release events?

Sub-questions

1. What are the allowable mechanical limits (acceleration, stress, deformation, etc.) for medical products during emergency release, and how should these define the design envelope?
2. How can the inflated airbag configuration be optimised to satisfy impact-absorption requirements while balancing competing design requirements?
3. How can the optimised inflated geometry be transformed into a compact and functionally deployable system?

## Design workflow (METHODOLOGY)

### 3.1. Problem statement

What I have been given, hypothesis of payload (Akineton Fiale, but most importantly the fact that they are simply exemplary phials) and payload bay (designed from former research) \* I focus on the inner design loop with safety, dictated by SORA

\* Dato che fino ad ora il problema è stato normativo, si è partiti da un approccio di calcolo della probabilità di failure. Ci sono state 5 tesi e svariati lavori scientifici con l'obiettivo generalizzato di quantificare il rischio di un'operazione di droni che sorvolano la città, per poi passare questo testimone a futuri ricercatori con l'obiettivo di MINIMIZZARLO (anche se alcune indicazioni sono state già effettivamente rilasciate)

1. Individuazione SCENARI DI MISSIONE - tesi BERTELLI 2021 \* Specific Operations Risk Assessment (SORA) - adottata da EASA, consiste in: \* Definizione dei Concept of Operations - dove abbiamo la DESCRIZIONE DELLA MISSIONE, che non si può cambiare \* Determinazione della Ground Risk Class (GRC) - ha un certo valore 6, pg. 39, per diminuirlo bisogna implementare metodi di MITIGAZIONE (3 tipologie), io agisco nell'ambito della mitigazione M2 (pg. 42, mitigare effetti dell'impatto al suolo). PG. 46!!!! M2, progettazione tecnica. Nuova valutazione del GRC pari a 3 (pg. 54) \* Valutazione dell'Air Risk Class (ARC) - collisione con altri mezzi nello spazio aereo, fuori dal mio scopo di tesi

\* Primatesta (ref. 8 e 9) - ispirazione anche per Davide \* L' Ospedale Molinette per l'AOU Città della Salute e delle Scienze e l'Ospedale San Giovanni Bosco per l'ASL Città di Torino - poli ospedalieri più fondamentali per la missione (posizionati ai lati opposti della città) (pg. 23) \* requisito del drone: deve essere già equipaggiato di paracadute (pg. 23)- failure del paracadute è lo scenario in cui mi colloco io. " Il paracadute può decelerare il carico utile fino ad una velocità di 3 m/s, al fine di fornire un atterraggio sicuro." pg.19 —> 3m/s causa comunque certe accelerazioni di impatto non indifferenti, che non garantiscono l'integrità dei medicinali. —> simula anche a 3m/s v verticale?!!?!?!?!? CASO ACQUA E CASO TERRA???

\* drone scelto: Matrice 300 RTK della casa produttrice DJI (che comunque deve essere equipaggiato di paracadute) - pg. 25 \* capacità massima di carico che equivale a 2,7 kg - LIMITE MASSIMO ENTRO CUI IL MIO PROGETTO (PARACADUTE, AIRBAG, CONTENITORE) DEVE STARE. \* Resistenza al vento: 15m/s in crociera, <https://www.dji-store.it/prodotto/dji-matrice-300-rtk/?srsltid=AfmBOOrTcxO1yqTrZgryaDUUjwIWj> <https://www.eliteconsulting.it/store/dji-matrice-300-rtk/>

\* la scelta del drone non è univoca e "Ed anzi, al momento della realizzazione effettiva del progetto, sarà prevista la progettazione di un UAV "ad hoc", che verrà progettato e realizzato col solo ed unico fine di condurre questa specifica missione" - pg. 25

\* pg 31 - "il drone mantenga durante tutta la tratta prestabilita una velocità di crociera di 17 m/s e che si muova ad una velocità pari a 5 m/s per effettuare la fase di decollo verticale fino alla quota di 75 m" \* fig 4, pg. 32 caso 1, fig. 5 pg. 33 caso 2 (leggermente più pericoloso, ma comunque) - tratta sul PO che

collega i due ospedali, 15 minuti di tragitto, 10km - paragonabile al tempo percorso su gomma (fattore traffico, anche se saremmo in ambulanza a sirene spiegate)

\* COLLEGAMENTO CON TESI DI DAVIDE - i suoi indici di rischio dicono che non è fattibile una missione con drone sul territorio cittadino —> quindi andiamo sul PO

\* possibili velocità di crociera:  $v = 17\text{m/s}$  realistica,  $23\text{m/s}$   $v$  massima realizzabile dal drone (pg 35)

\* Altezza operativa massima prevista: 75 m. \* MTOM ("Maximum Take Off Mass") del drone: 9 kg. \* Dimensione massima caratteristica (d) dell'aeromobile: 0.81 m. \* Velocità (v): 17 m/s.

\* tabella 8 pag. 67

\* terminal velocity calculator that I can use too: <https://www1.grc.nasa.gov/beginners-guide-to-aeronautics/termvel/>

\* rischi legati all'ottimizzazione dell'ultima tratta su terra (vari scenari: linea retta, un parco, etc.) — da considerare?

TESI CAVALLO 2020 \* Individuazione tratta Molinette - San Giovanni Bosco (pg. 15) \* Distinction between two types of transportation: Hospital-to-Hospital, Hospital-to-Home (pg. 20)—> my case is the first one. \* Wind information: the city of \* idea originale del corridoio acquatico soddisfa i requisiti normativi restringenti dell'enac (pag. 37)

Caratteristiche del drone: \* Cruise speed in m/s; —> 70 • Maximum flight time in minutes; —> half of drone's autonomy • Mass in kg; • Radius in m; • Frontal Area in m<sup>2</sup>; • Glide speed in m/s; —> calculated differently for each drone type ( pg 43) \* WIND —> average of 2m/s throughout the year, 23m/s max.

\* two types of descent: ballistic descent and uncontrolled glide are already mentioned here \* first to suggest the river corridor solution —> brings to Bertelli

2. CONTENITORE DA PORTARE - trasporto dei medicinali, tesi SIVIERI 2021 \* pg 19 sivieri: "Il tronco centrale è a sezione cilindrica costituito da 4 strati. Partendo dallo strato interno e procedendo a quello più esterno: 1. Strato interno di materiale PETG di spessore 2 mm; 2. Strato di materiale isolante di natura e spessore incognito; 3. Strato esterno di materiale PETG di spessore 2 mm; 4. Strato di vernice termo riflettente. • Le estremità sono costituite cupole emisferiche di costruzione equivalente a quella del tronco. \* progettato per soddisfare requisiti termici e vibrazionali di una missione tipica \* CARATTERISTICHE (pg. 27 - 31 SIVIERI) \* messa in tavola fig. 2.12 \* spessore di 2 mm sulla parete esterna ed interna, con un riempimento interno dell' 1 % con pattern rettangolare, come in 2.15 \* la cavità interna è stata riempita di schiuma poliuretanica (scelta di iniettabilità efficace) \* superficie esterna è stata scartavetrata e sono state applicate tre strati di vernice argentata (2.16) per migliorare la riflettanza del materiale

\* MEDICINALI DA TRASPORTARE - PARENTESI SULLE TIPOLOGIE DI CONTENITORI E SCATOLE DI FARMACI QUI

3. INDICI DI MERITO BASATI SUL RISCHIO - TESI ANGELINI 2023: \* Metodo per determinare la traiettoria di rischio minore: determinare la traiettoria di minimo rischio tra due punti si usa l'algoritmo di Dijkstra e l'algoritmo A\* (del quale Dijkstra è un caso particolare). pg 17 \* Mappe di input: densità della popolazione (pg. 35), mappa dei "ripatri" (ref. 17 nella tesi, pg. 40) cioè degli edifici che rappresentano un fattore di sicurezza per il sorvolo della città (stima di vari fattori di protezione ispirati a diversi lavori scientifici)

\* modi di caduta più frequenti: ballistic descent (forza di gravità e resistenza aerodinamica, eq. 3.3 - risolta numericamente per trovare velocità e punto di impatto al suolo, tabella 3.3) and uncontrolled glide (ref 17 in tesi) —> ballistic descent: velocità  $x$  iniziale VTOL = 140m/s, velocità del vento media 25.2 m/s, —> uncontrolled glide: \* tempo di impatto al suolo eq. 3.11 \* distanza percorsa eq. 3.10 \* mappe di rischio generate con metodi probabilistici \* CALCOLO DEL PERCORSO OTTIMO focus drone: —> traiettoria di minimo rischio con l'algoritmo Dijkstra applicato alla mappa di rischio generata Funzionamento: "algoritmo statico che calcola il percorso di minimo rischio su una mappa in cui per ogni cella c'è un livello di rischio associato." —> risultato: mappe sulla provincia di torino ed oltre con percorso a minimo rischio. \* ref. 8 il lavoro di Romeo e Cestino \* INDICE DI MERITO: \*

3 condizioni: payload, lunghezza del tragitto, sicurezza (nel caso del drone molto più penalizzante)  
\* calcolo dell'indice di merito cambia dipendentemente dalla tipologia di missione (urgente, media, routine), pg. 86

\* PRECEDENTE AD ANGELINI - TESI FIORENTINO 2021 \* path-planning based on RRT\* Method \* concept of "layers" as in factors that need to be combined to have a full picture of the risk analysis \* primatesta cited again (pg 40 explanation)

### **3.2. (Outer) design loop formalisation**

Aeronautical-projects inspired design loop WITH shift on payload-centric design, "protect the cargo rather than the machine" (rather new for the aerospace sector, where cargo and plane are NOT handled separately like in this case). Idea of the airbag.

### **3.3. Proposed approach**

Inner design loop formalisation: specific requirements and design steps The diagram + objectives, i.e. where this MSc Thesis "stops" (slide 8 of Mid-term meeting pptx)

# 4

## Testing the Design Worklow RESULTS

Do not include too many implementation details

### 4.1. Mission and failure scenario analysis: calculating $V_x$ and $V_y$

\* Mi servono degli input: prendo le informazioni raccolte nella fase preliminare di analisi missione (ref analisi missione) per stabilire probabilità di impatto and what not.

\* traiettoria \* punto di partenza, punto di arrivo \* drone scelto \* payload scelto \* medicinali scelti per simulazione

\* serve energia di impatto?

—> risultati del WP2

### 4.2. Studying the payload: calculating allowable impact acceleration

#### 4.2.1. Material characterisation

TEST

FEM

MENTION MODEL ITERATIONS TRIAL-AND-ERROR IN THE APPENDIX A!!!! FROM ABAQUS IMPLICIT TO CURRENT DECISIONS. Step by step, as in "version 1, abaqus implicit". Version 2, ls-dyna tetra elastic. Etc. MODAL PERTURBATION

\* impact of container in LS-Dyna and what not. \* diversi angoli di assetto per vedere il caso peggiore di accelerazioni locali \* cosa possono sopportare le fiale e il contenitore???? \* tipi di impatto, contenitore tipico che le contiene (menzionare la tesi del tipello) e che per il dimensionamento non viene considerato, approssimato a infinitamente rigido e che trasmette le accelerazioni di impatto così come le riceve —> airbag deve essere dimensionato così —> tutte le analisi statistiche di prima, che hanno tenuto in considerazione anche il fatto che prima o poi si sorvolerà la città una volta risolti i cavilli burocratici che impongono una certa mitigazione del rischio —> airbag.

—> risultati del WP3

—> CONCLUSIONE: I have what I need to design the damn airbag.

#### 4.2.2. Sled test

### 4.3. Design variables optimisation: the code!!!

\* codice ispirato da Zhou et al 2019 - spiegazione del modello e dell'aspetto dei vari blocchi \* Sapendo che le thresholds delle provette sono tot, allora si può convergere verso un pre-dimensionamento dell'airbag

con tot dimensioni o caratteristiche. \* oracle code for direzionamento verso un punto di partenza per il design

\* I CHOSE THE CRUDE IMPLEMENTATION — THIS IS BECAUSE THE ONLY COUPLING ANALYTICAL MODEL IS MISSING AND I CANNOT RETRIEVE IT. THEREFORE THIS IS MY IMPLEMENTATION XXX Title: "Design, Fabrication and Testing of an Airdrop Platform Utilizing Airbags as Shock Absorbers"

Author: Timothy F. Patterson

Report: NATICK/TR-87/042 (DTIC accession ADA199593)

Date: August 1985 (work performed March 1984–May 1985)

## **4.4. Summary of results**

Actual design variables p inflation, diameter, shape, etc

**\*\*RESULTS: BRING IT BACK TO THE RESEARCH QUESTIONS!!!!\*\***

# 5

## Conclusions

### **5.1. General summary**

Bring back the idea of the safety-first workflow into the broader context of the long-term, drone-network-all-over-Italy vision.

### **5.2. Recommendations for future work**

Feeding these results in the “outer” design loop, updating the hardware and mission design, re-evaluating initial conditions to re-inform the “inner” design loop -> once reached convergence, the need to simulate and test the design implemented on the drone, with informed decisions on inflation mechanism and folding method.

# 6

## Project Management

Inspired by the Gibb's reflective cycle (?)

### 6.1. Strategy and planning

planning strategy (especially at the beginning), 5 supervisors to update independently, different means of updates - weekly meetings that became only when i had updates - regular bi-weekly meetings - irregular meetings and informal discussions here and there because of huge workload

### 6.2. Challenges

#### 6.2.1. challenge 1: aligning interests for thesis scope

- 5 supervisors with 5 different visions of my thesis work. what contributed was a lack of a unified platform to check my progress and congregate materials.

#### 6.2.2. challenge 2: Computational resources NO MORE

- by contract, politecnico should have given the resources, hpc and licences - unexpected maintenance, with delay - Bianca Ex Machina, access to DelftBlue. Out of contract, otherwise the thesis should have been changed

#### 6.2.3. challenge 3: IT infrastructure bottleneck

- resulting project flow from company pc to personal pc, because no permission to install a VPN and launch from the company's computer.

#### 6.2.4. challenge 4: going against the judgement of supervisors

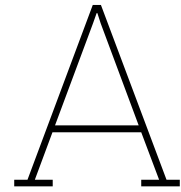
- scegliere shell invece che solids in case of fev supervisor - scegliere full invece che reduced model in case of tudelft supervisor.

### 6.3. Lessons learned

I'm really mad

# References

- [1] G. Fantato et al., "Italian territorial emergency medical services - 118: A type of advanced EMS vehicle in the experience of the Piedmont region," *Ital. J. Emerg. Med.*, vol. 12, no. 2, pp. 81–89, Aug. 2023. doi: 10.23736/S2532-1285.23.00185-4
- [2] N. Abrescia, A. Maddaloni, G. Molinaro, A. Capponi, and M. D'Abbraccio, "The long crisis of Italy's National Health Service: Universalism on paper, inequity in practice," *Eur. J. Public Health*, vol. 36, no. 4, ckag138, Aug. 2026. doi: 10.1093/eurpub/ckag138
- [3] L. Garattini, M. Zanetti, and N. Freemantle, "The Italian NHS: What lessons to draw from COVID-19?" *Appl. Health Econ. Health Policy*, vol. 18, no. 4, pp. 463–466, Aug. 2020. doi: 10.1007/s40258-020-00594-5
- [4] M. Capuzzo, G. L. Viganò, C. Boniotti, L. M. Ignotti, C. Duri, and V. Cimolin, "Impact of the first phase of the COVID-19 pandemic on the acquisition of goods and services in the Italian health system," *Int. J. Environ. Res. Public Health*, vol. 19, no. 4, p. 2000, Feb. 2022. doi: 10.3390/ijerph19042000
- [5] M. Balasingam, "Drones in medicine—The rise of the machines," *Int. J. Clin. Pract.*, vol. 71, no. 9, e12989, Sep. 2017. doi: 10.1111/ijcp.12989
- [6] J. E. Scott and C. H. Scott, "Drone delivery models for healthcare," in *Proc. 50th Hawaii Int. Conf. Syst. Sci. (HICSS)*, Jan. 2017, pp. 3297–3304. doi: 10.24251/hicss.2017.399
- [7] E. B. Odigie, B. I. G. Adejumo, and A. E. Oigbochie, "Importance of drones in healthcare delivery amid a pandemic: Current and generation next application," *Open J. Med. Res.*, vol. 2, no. 1, pp. 1–13, 2021. doi: 10.52417/ojmr.v2i1.187
- [8] A. Gangwal, A. Jain, and S. Mohanta, "Blood delivery by drones: A case study on Zipline," *Int. J. Innov. Res. Sci. Eng. Technol.*, vol. 8, no. 8, pp. 8760–8766, Aug. 2019. doi: 10.15680/IJIRSET.2019.0808063
- [9] F. Cavallo, "Innovative ways of distribution for life-saving medicines during emergency situations," Master's Thesis, Politecnico di Torino, Torino, Italy, 2020.
- [10] D. Bertelli, "Studio di fattibilità di un sistema innovativo, tramite drone, per il trasporto di farmaci salvavita in situazioni di emergenza," Master's Thesis, Politecnico di Torino, Torino, Italy, 2021.
- [11] F. Sivieri, "Valutazione degli effetti ambientali sul trasporto di materiale biologico e farmaci tramite aeromobili a pilotaggio remoto," Master's Thesis, Politecnico di Torino, Torino, Italy, 2021.
- [12] F. Fiorentino, "Path planning strategies for drone delivery for life-saving pharmaceuticals," Master's Thesis, Politecnico di Torino, Torino, Italy, 2021.
- [13] D. Angelini, "Progettazione e valutazione di un sistema di trasporto innovativo aereo per farmaci emoderivati in condizione di emergenza," Master's Thesis, Politecnico di Torino, Torino, Italy, 2023.
- [14] D. Angelini, E. Cestino, D. Cestino, and F. Cattel, "Comparative analysis of evtol, drone and ground transportation systems for emergency delivery of blood-derived medication," in *Proceedings of the International Council of the Aeronautical Sciences (ICAS)*, Florence, Italy: ICAS, 2024.



## FEM Iterations for glass phial model

*Adding source code to your report/thesis is supported with the package listings. An example can be found below. Files can be added using \lstinputlisting[language=<language>]{<filename>}.*

```
1 """
2 ISA_Calculator: import the function, specify the height and it will return a
3 list in the following format: [Temperature, Density, Pressure, Speed of Sound].
4 Note that there is no check to see if the maximum altitude is reached.
5 """
6
7 import math
8 g0 = 9.80665
9 R = 287.0
10 layer1 = [0, 288.15, 101325.0]
11 alt = [0, 11000, 20000, 32000, 47000, 51000, 71000, 86000]
12 a = [-.0065, 0, .0010, .0028, 0, -.0028, -.0020]
13
14 def atmosphere(h):
15     for i in range(0, len(alt)-1):
16         if h >= alt[i]:
17             layer0 = layer1[:]
18             layer1[0] = min(h, alt[i+1])
19             if a[i] != 0:
20                 layer1[1] = layer0[1] + a[i]*(layer1[0]-layer0[0])
21                 layer1[2] = layer0[2] * (layer1[1]/layer0[1])**(-g0/(a[i]*R))
22             else:
23                 layer1[2] = layer0[2]*math.exp((-g0/(R*layer1[1]))*(layer1[0]-layer0[0]))
24     return [layer1[1], layer1[2]/(R*layer1[1]), layer1[2], math.sqrt(1.4*R*layer1[1])]
```

# B

## Task Division Example

*If a task division is required, a simple template can be found below for convenience. Feel free to use, adapt or completely remove.*

**Table B.1:** Distribution of the workload

| Task                       |              | Student Name(s) |
|----------------------------|--------------|-----------------|
|                            | Summary      |                 |
| Chapter 1                  | Introduction |                 |
| Chapter 2                  |              |                 |
| Chapter 3                  |              |                 |
| Chapter *                  |              |                 |
| Chapter *                  | Conclusion   |                 |
| Editors                    |              |                 |
| CAD and Figures            |              |                 |
| Document Design and Layout |              |                 |